Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>De Broglie wavelength formula: Î» = h / (p)

Where h = Planck's constant, ~6.625 ÃĹ10^-34 Js

p = momentum = mv

To find momentum:

K = (1/2)mv^2

E = 1eV = 1.602 ÃĹ10^-19 J

At 1 eV energy,

m*v^2 = (2*K) = (2*1.602 ÃĹ10^-19)

v^2 = 3.204 ÃĹ10^-19

v = 5.661 ÃĹ10^4 m/s

p = m*v = 9.104 ÃĹ10^-31 * 5.661 ÃĹ10^4 = 5.135 ÃĹ10^-26 kgÂ·m/s

Î» = h / (p) = 6.625 ÃĹ10^-34 / 5.135 ÃĹ10^-26 = 1.293 ÃĹ10^-8 m

Convert to cm: 1.293 ÃĹ10^-8 * 10^4 = 1.293 ÃĹ10^(-4) cm

Rounding to the nearest option, it is 1.23 ÃĹ10^-7 cm. </think>

<answer>1.23 ÃĹ10^-7 cm </answer>